

PREPARED FOR

Ashesi University — Classroom Booking System Project
Contact: Project Stakeholder / Booking System Lead

Project Summary

Recent delivery covered production bug fixes, feature enhancements to the room-search booking flow, UI/UX polish, and comprehensive architecture documentation updates for the AURA platform. This quote itemises the engineering effort, tooling, quality assurance, security review, and platforms delivered.

1. Engineering Effort

Work package	Rate (GHS)	Hours	Amount (GHS)
Calendar dayBoundaries fix Schedule-X whole-hour validation, grid height tuning, label updates	500	6	3,000
Booking dialog enhancement Editable room/date/start/end, availability pre-check, capacity validation	500	22	11,000
UI/UX fixes Uniform button radius, overlay z-index fixes for popovers/selects/tooltips	500	8	4,000
Code review & bug fixes PENDING-booking logic correction, error handling, cross-module review	500	10	5,000
Architecture documentation New core-algorithms section, PDF regeneration	500	12	6,000
Testing, QA & typecheck Vitest unit tests, TypeScript checks, integration validation	500	9	4,500
Deployment & release management Git commits, pushes to main, Vercel-ready release coordination	500	5	2,500
Subtotal — engineering effort			38,000

2. Fixed Deliverables

Deliverable	Amount (GHS)
Security & privacy review (RBAC, JWT, CSRF, audit, Ghana Act 843 alignment)	6,500
Tooling & platform enablement (Vercel + Render pipeline, GitHub workflow)	4,000

Deliverable	Amount (GHS)
Documentation package (ARCHITECTURE.md + PDF, commit messages, change notes)	3,500
Subtotal — fixed deliverables	14,000
Total before tax	52,000
VAT / NHIL / GETFund (estimated 22.5%)	11,700
GRAND TOTAL	GHS 63,700

3. Tools & Technologies Used

Kimi Code CLI	Next.js 16	React 19	TypeScript	Go	PostgreSQL 18	Tailwind CSS v4
Radix UI	Schedule-X	React Hook Form	TanStack Query	sqlc	Vitest	Playwright
Git / GitHub	Vercel	Render	Pandoc	WeasyPrint		

4. Platforms & Languages

Platforms delivered • Web requester portal (Next.js App Router) • Admin console (RBAC-gated) • Mobile app (Expo SDK 56 / NativeWind) • REST API (Go / chi router) • Cloud deploy: Vercel + Render Blueprint	Languages & data • TypeScript / JavaScript (web + mobile) • Go (backend services) • SQL (PostgreSQL migrations & queries) • CSS / Tailwind (design system) • OpenAPI 3.1 (contract-first clients)
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5. Security, Privacy & Quality

Security • Argon2id password hashing • JWT access + opaque refresh tokens • httpOnly / __Host- cookies • Double-submit CSRF protection • RBAC permission matrix • Rate limiting & security headers • Append-only audit log	Privacy & compliance • Ghana Data Protection Act 843 alignment • Privacy/terms pages (public route) • User consent & device-token handling • Role-scoped data access • Audit trail on every booking state change
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Quality gates run: TypeScript type-checking, Go vet, Vitest unit tests (74 passed), integration validation, manual cross-browser checks, and production-deploy readiness review.

6. Features Available on the Platform

Room search & availability grid	Booking request & approval workflow	Unified calendar (day/week/month)
Maintenance windows	Timetable CSV/XLSX import	Notifications (SSE + email + push)
Utilisation & conflict reports	Admin override & audit log	User/department/role management
Semester & room catalogue CRUD		